CS 59300 Application of Deep Learning

Homework 5: Kaggle Competition

Due on **October 8, Wednesday, 2025**, 11:59pm

The goal of this homework assignment is to help you apply what you learned in Machine Learning (ML) to a Kaggle competition. Please refer to the following link at Kaggle for "Titanic – Machine Learning from Disaster":

https://www.kaggle.com/competitions/titanic

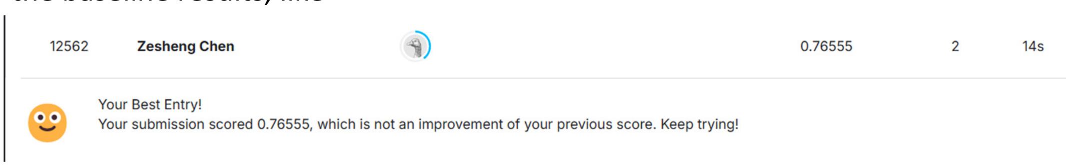
Please refer to the following link about Titanic Tutorial for how to submit your results at Kaggle:

https://www.kaggle.com/code/alexisbcook/titanic-tutorial

Note that following the above tutorial would give you a score of **0.77511**.

Please follow the following steps:

(1) Provide a **baseline** for this classification problem. Name the Jupyter Notebook as "**baseline.ipynb**". Please evaluate the performance of the baseline, including accuracy, confusion matrix, and F1 score, on the training dataset. Please make sure that the score is **at least 0.5**. Take the screenshot of you current ranking at Leaderboard after submitting the baseline results, like



and put this **screenshot** and **performance evaluation results** in a doc file. The above screenshot showed that my baseline has a score of 0.76555.

(2) Design a machine learning method that can beat the baseline and has a score **more than 0.784**. Please refer to the end-to-end machine learning project example covered in the class (i.e., predict median house values in Californian districts) or https://github.com/ageron/handson-ml3/blob/main/02_end_to_end_machine_learning_project.ipynb
Please apply the techniques, such as data preprocessing and machine learning models, to get a better score at Kaggle. Name the Jupyter Notebook as "**machine_learning.ipynb**", and take a screenshot of your ranking at Leaderboard after applying machine learning methods or techniques and put it in the doc file.

Please answer the following questions for the **final** machine learning techniques you apply in the doc file:

1. What data preprocessing methods did you apply, such as feature scaling with a min-max method or one-hot encoding for a feature?
2. What machine learning model(s) did you apply, such as SVM or random forest?
3. What are performance evaluation results on training dataset, including accuracy, confusion matrix, and F1 score, on the training dataset?
4. What the best score and ranking at Leaderboard at Kaggle did you receive?

Note: Please do not refer to any existing solution/code or use AI tools, except Titanic Tutorial at https://www.kaggle.com/code/alexisbcook/titanic-tutorial. That is, you cannot refer to any existing solution referred at https://www.kaggle.com/competitions/titanic and other places, except Titanic Tutorial at https://www.kaggle.com/code/alexisbcook/titanic-tutorial. Basically, you will come up with your own implementation on this Kaggle competition. If you violate this policy, you will get 0 for this homework assignment and receive a warning! Please follow your honor code!

Note: Please do not discuss with each other.

Note: Please apply only datasets at https://www.kaggle.com/code/alexisbcook/titanic-tutorial/input, not any other datasets.

Note: Each day you can submit only 10 times at Kaggle. Please submit to competition when you have a major or essential change to the code.

Please create "**hw5**" folder in the root of your GitHub homework repository and then upload your Jupyter Notebooks (i.e., baseline.ipynb and machine_learning.ipynb) and a doc file that contains the screenshots and your answers to the above questions to be under "hw5" folder at GitHub.

Grading rubric:

- Baseline and performance evaluation of the baseline method: 20 pts
- Your machine learning implementation and answers to questions: 100 pts
  - If score > **0.784**, would get 100 pts
  - Else if score > 0.783, would get 90 pts
  - Else if score > 0.782, would get 80 pts
  - …
  - Else if score > 0.776, would get 20 pts

    - Else if score > 0.775, would get 10 pts
    - Else, would get 0.

- (**Bonus**) **Top three** students among this CS59300 class listed at the Leaderboard at Kaggle: 30 pts
- **Note**: If you do not answer the questions, you lose half of points from what you get based on the performance score.

Total: 120 pts